A5 Convolutional Neural Networks

Artificial Intelligence
AIN2IL (MTD.ba)
SS2026

Submission via Moodle **30 Points**

Exercise 1 (30 points) The hands-on workshop for CNN was devoted to training models for recognizing 10 classes from the CIFAR-10 dataset. Start from the notebook `CNN_CIFAR10.ipynb`. Train three different models (you may, but need not, restrict to 10 epochs for each training run). Here is a list of examples that you might find useful to change:

- Add/remove convolutional layers
- Change the numbers of filters/neuron of the convolutional layers
- Change the number of neurons in the last hidden layer
- Either use `Flatten()` or `GlobalAveragePooling2D()` to perform the transition from the convolutional network to the fully connected network

Make sure that the definitions and training runs of all three networks are included in the notebook (please make copies of the sections “Create Model #1”, “Train Model #1”, “Evaluate Model #1 on Test Data (best model)” for each model you train). Select the best model in terms of validation accuracy (the one that had been saved last in the training run).

Warning: Be aware that one epoch may take several minutes if a GPU is used (and much more if no GPU is used). So give yourself sufficient time and start working on the exercise soon enough!

Please hand in the following:

1. *The best model under the exact file name `Surname_FirstName.keras` (insert your name; no special characters! upload link provided in class)*
2. *Your notebook and an HTML dump of it.*

The best three models will receive three extra points.